

MATH 341: Homework 4

Aren Vista

March 5, 2026

1 Question 1

What lower triangular matrix E puts A into upper triangular form $EA = U$?

Multiply by $E^{-1} = L$ to factor A into LU :

$$\begin{aligned}
 A &= \begin{bmatrix} 2 & 1 & 0 \\ 0 & 4 & 2 \\ 6 & 3 & 5 \end{bmatrix} \\
 \underbrace{\begin{bmatrix} 2 & 1 & 0 \\ 0 & 4 & 2 \\ 6 & 3 & 5 \end{bmatrix}}_A &\xrightarrow{R_3=R_3-3R_1} \underbrace{\begin{bmatrix} 2 & 1 & 0 \\ 0 & 4 & 2 \\ 0 & 0 & 5 \end{bmatrix}}_U \implies \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix}}_E \\
 \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix}}_E &\xrightarrow{\text{Multiply off-diag } -1} \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 3 & 0 & 1 \end{bmatrix}}_L \\
 \begin{bmatrix} 2 & 1 & 0 \\ 0 & 4 & 2 \\ 6 & 3 & 5 \end{bmatrix} &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 3 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 & 0 \\ 0 & 4 & 2 \\ 0 & 0 & 5 \end{bmatrix}
 \end{aligned}$$

$$A = LU$$

2 Question 2

This problem shows how the one-step inverses multiply to give L . You see this best when $A = I$ is already lower triangular with 1's on the diagonal. Then $U = I$

$$A = \begin{bmatrix} 1 & 0 & 0 \\ a & 1 & 0 \\ b & c & 1 \end{bmatrix} \quad E_1 = \begin{bmatrix} 1 & 0 & 0 \\ -a & 1 & 0 \\ -b & 0 & 1 \end{bmatrix} \quad E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -c & 1 \end{bmatrix}$$

- 1) E_2E_1 to find $EA = I$
- 2) $E_1^{-1}E_2^{-1}$ to find $A = L$
- 3) The multipliers a, b, c are mixed up in $E = L^{-1}$ but are perfect in L

2.1 P.1

$$\begin{bmatrix} 1 & 0 & 0 \\ -a & 1 & 0 \\ -b & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -c & 1 \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ -a & 1 & 0 \\ -b & -c & 1 \end{bmatrix}}_{E_1 E_2}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ -a & 1 & 0 \\ -b & -c & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ a & 1 & 0 \\ b & c & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

2.2 P.2

$$E_1^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ a & 1 & 0 \\ b & 0 & 1 \end{bmatrix} \quad E_2^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & c & 1 \end{bmatrix} \quad E_1^{-1} E_2^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ a & 1 & 0 \\ b & c & 1 \end{bmatrix} = L$$

3 Question 3

Compute L, U for the matrix A :

$$A = \begin{bmatrix} a & a & a & a \\ a & b & b & b \\ a & b & c & c \\ a & b & c & d \end{bmatrix}$$

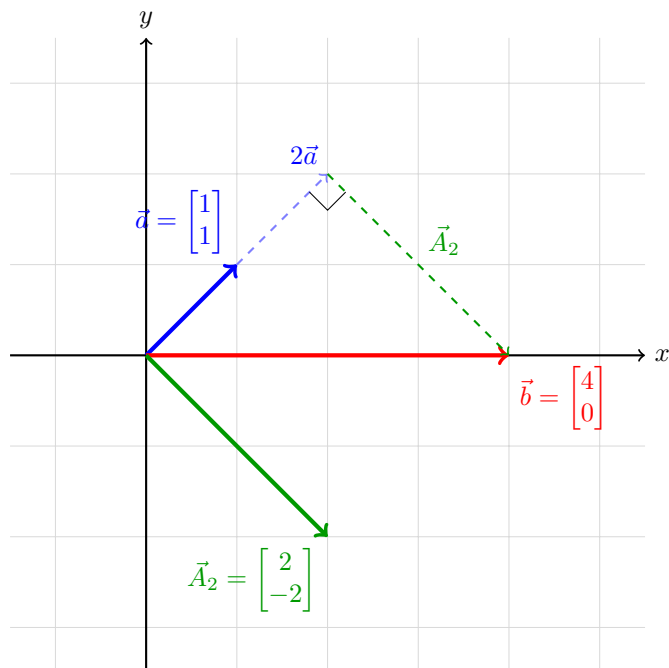
Find four condition on a, b, c, d to get $A = LU$ with four non-zero pivots.

4 Question 4

What multiple of a should be substraced from b to make the resut A_2 orthogonal to a ? Sketch a figure to show a, b, A_2

$$a = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad b = \begin{bmatrix} 4 \\ 0 \end{bmatrix}$$

$$\begin{aligned} \exists A_2 = b - xa \ni A_2 \perp a \\ \implies a \cdot A_2 = 0 & \implies a \cdot (b - xa) = 0 & \implies ab - xaa = 0 \\ \implies x = \frac{ab}{aa} = \frac{(1 \times 4) + (0)}{2} = \frac{4}{2} = 2 \\ A_2 = b - 2a = \begin{bmatrix} 2 \\ -2 \end{bmatrix} \end{aligned}$$



5 Question 5

Complete the Gram-Schmidt process in the prior problem via computing

$$q_1 = \frac{a}{\|a\|} \quad A_2 = b - (b^T q_1) q_1 \quad q_2 = \frac{A_2}{\|A_2\|}$$

$$a = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad b = \begin{bmatrix} 4 \\ 0 \end{bmatrix} \quad A_2 = \begin{bmatrix} 2 \\ -2 \end{bmatrix}$$

$$\|a\| = \sqrt{1^2 + 1^2} = \sqrt{2} \implies q_1 = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$b^T q_1 = [4 \quad 0] \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} = \frac{4}{\sqrt{2}}$$

$$\frac{4}{\sqrt{2}} \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$

$$\begin{bmatrix} 4 \\ 0 \end{bmatrix} - \begin{bmatrix} 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ -2 \end{bmatrix} = A_2$$

$$\|A_2\| = \sqrt{4 + 4} = 2\sqrt{2}$$

$$q_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 2 \\ -2 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix}$$